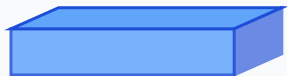


1. Multi-Scale Backbone & FPN

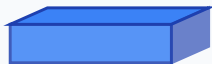
Input Image $I \in \mathbb{R}^{H \times W \times 3}$
(Aerial / Maritime Scenes)



P_2 ($H/4 \times W/4$)



P_3 ($H/8 \times W/8$)



P_4 ($H/16 \times W/16$)



P_5 ($H/32 \times W/32$)

ResNet-50 + FPN
Shared Feature Pyramids

2. Stage 1: RPN Homotopy Assignment

$$S_{\text{H-WIoU}}(\mathbf{A}, \mathbf{B}) = [\text{IoU}(\mathbf{A}, \mathbf{B})]^{\gamma(s_B)} \cdot e^{-(1 - \gamma(s_B))D_W^2(\mathbf{A}, \mathbf{B})}$$

- Dynamic Top-k Candidate Selection
- Continuous IoU $\leftrightarrow$ Wasserstein Interpolation
- Positive Anchor Survival Rate: 0.18 $\rightarrow$ 0.94

4. Scale Homotopy Modulation Engine

$$\gamma(s) = \frac{s^2}{s^2 + \sigma_0^2}, \quad s = \sqrt{w \cdot h}$$

- $s \rightarrow 0$ (Microscopic)** $\rightarrow \gamma(s) \rightarrow 0$
 $\rightarrow$ Optimal Transport $\mathcal{W}_2$ (No Vanishing Gradients)
- $s \gg \sigma_0$ (Standard)** $\rightarrow \gamma(s) \rightarrow 1$
 $\rightarrow$ Discrete Lebesgue Measure IoU (High AP₇₅)

Zero Parameter Bloat (+0 MB), Real-Time (54.5 FPS)

3. Stage 2: RoI Head & Homotopy Loss

RoIAlign Pooling (7times7)
Spatial Feature Extraction

Classification Head
 $\mathcal{L}_{\text{cls}} = \text{CrossEntropy}(\hat{p}, y)$

Homotopy Bounding Box Loss
 $\mathcal{L}_{\text{H-WIoU}} = 1 - S_{\text{H-WIoU}}(\mathbf{P}_i, \mathbf{G}_i)$

- Bounded Gradient: $\lim_{s \rightarrow 0} \|\nabla \mathcal{L}\| = \mathcal{O}(1)$
- Strict Boundary Precision: $\lim_{s \rightarrow \infty} \mathcal{L} = \mathcal{L}_{\text{IoU}}$